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Privacy-Aware IPv6 IoT Security Monitoring Across Address Rotation

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ABSTRACT & PROBLEM DESCRIPTION

Abstract

IPv6 privacy extensions allow IoT devices to rotate temporary addresses, reducing address-based tracking but complicating local security monitoring. Priv6Guard proposes a privacy-aware monitoring system that correlates security events through authenticated, short-lived local sessions and pseudonyms rather than permanent device identifiers. The system monitors IPv6 traffic, detects suspicious Router Advertisements and abnormal Neighbor Discovery Protocol (NDP) activity, and presents alerts through a web dashboard. A controlled five-device IPv6 LAN will be used to evaluate detection performance, session continuity, pseudonym expiration, and cross-tenant isolation.

Problem Description

- Temporary IPv6 addresses make address-based monitoring less reliable.
- Rogue Router Advertisements can introduce unauthorized IPv6 configuration.
- Excessive NDP messages may indicate suspicious or malicious activity.
- Traditional persistent identifiers can create privacy and tracking risks.
- Security monitoring must continue across address rotation without reconstructing permanent device identities.

Background

IPv6 uses Neighbor Discovery Protocol (NDP) for neighbor discovery, address resolution, router discovery, and reachability management. Router Advertisements (RAs) provide hosts with router and network configuration information.

RFC 8981 describes temporary IPv6 addresses with randomized interface identifiers to limit address-based activity correlation. However, changing addresses can make it harder for monitoring systems to associate related security events.

Key Concepts

- **IPv6 Privacy Extensions:** Temporary addresses help limit long-term address-based tracking.
- **Neighbor Discovery Protocol:** Supports communication and router discovery on IPv6 links.
- **Rogue Router Advertisement:** An unauthorized or suspicious RA may manipulate host configuration.
- **NDP Rate Anomaly:** An unusually high rate of Neighbor Discovery messages may indicate abuse.
- **Short-Lived Pseudonyms:** Temporary local identifiers used for limited-duration security correlation.

Proposed Optimization

Separate network addressing from authenticated session management. Correlate events only within an authorized local scope, expire pseudonym mappings automatically, and avoid behavioural fingerprinting or permanent identity recovery.

Literature & RFC Review

Relevant Standards

1. RFC 8981 — Temporary Address Extensions for Stateless Address Autoconfiguration in IPv6

Defines temporary IPv6 addresses with randomized interface identifiers to reduce the period during which address-based tracking is straightforward.

Relevance: Establishes the privacy behaviour that Priv6Guard must accommodate.

2. RFC 4861 — Neighbor Discovery for IP Version 6 (IPv6)

Defines IPv6 Neighbor Discovery, including Router Solicitation, Router Advertisement, Neighbor Solicitation, and Neighbor Advertisement messages.

Relevance: Provides the protocol foundation for packet classification and NDP monitoring.

3. RFC 9099 — Operational Security Considerations for IPv6 Networks

Discusses IPv6 operational security challenges and mitigation considerations for managed networks.

Relevance: Supports the security-monitoring and deployment design.

4. RFC 4890 — Recommendations for Filtering ICMPv6 Messages in Firewalls

Explains considerations for filtering ICMPv6 while preserving messages required for correct IPv6 operation.

Relevance: Highlights why detection and mitigation must account for legitimate ICMPv6 traffic rather than indiscriminately blocking it.

Identified Project Gap

Priv6Guard addresses the combined requirement of short-lived authenticated local correlation, address-rotation-aware monitoring, suspicious RA detection, NDP-rate anomaly detection, and explicit privacy validation.

Note: This is the project's design objective, not a claim that the cited RFCs prescribe this complete monitoring architecture.

Proposed Solution

Priv6Guard: Privacy-Aware IPv6 Security Monitoring

Priv6Guard combines IPv6 traffic monitoring, temporary pseudonym management, authenticated sessions, rule-based threat detection, and a web dashboard.

1. IPv6 Test Environment

- One isolated IPv6 LAN with five emulated IoT devices.
- Controlled temporary-address rotation.
- A separate test actor for authorized attack simulation.

2. Authentication & Pseudonym Manager

- Authenticate enrolled device sessions.
- Generate short-lived, scope-specific pseudonyms.
- Associate permitted address changes with an active authenticated session.
- Expire mappings and delete them when their lifetime ends.

3. Network Monitoring

- Capture and classify IPv6/ICMPv6 traffic.
- Identify RA, RS, NS, and NA messages.
- Measure NDP message rates over configurable time windows.

4. Detection Engine

- Flag RAs that violate the configured local router trust policy.
- Detect abnormal NDP message rates using configurable thresholds.
- Generate alerts with severity, event type, time, and temporary local context.

5. Dashboard & Evaluation

- Display sessions, address rotations, pseudonym expiry, traffic rates, and alerts.
- Report detection rate, false-positive rate, and detection latency.
- Test tenant isolation and privacy-related limitations.

Expected Outcome

A working prototype that demonstrates address-rotation-aware local security monitoring without relying on permanent device identifiers or behavioural fingerprinting.

Optimization Proposed

Aspect	Conventional Approach	Priv6Guard Approach
Device Tracking	Relies heavily on IP address continuity	Uses temporary authenticated sessions
Address Rotation	May fragment monitoring records	Support session-scoped correlation
Device Identifiers	May retain persistent identifiers	Uses short-lived local pseudonyms
RA Monitoring	Manual inspection or basic filtering	Configurable trust-policy checks and alerts
NDP Monitoring	Limited rate visibility	Window-based rate monitoring
Alert Presentation	Raw packet logs	Dashboard with event details and trends
Privacy Controls	Retention may be indefinite	Expiry and deletion of pseudonym mappings
Evaluation	Qualitative demonstration	Measured detection and false-positive metrics

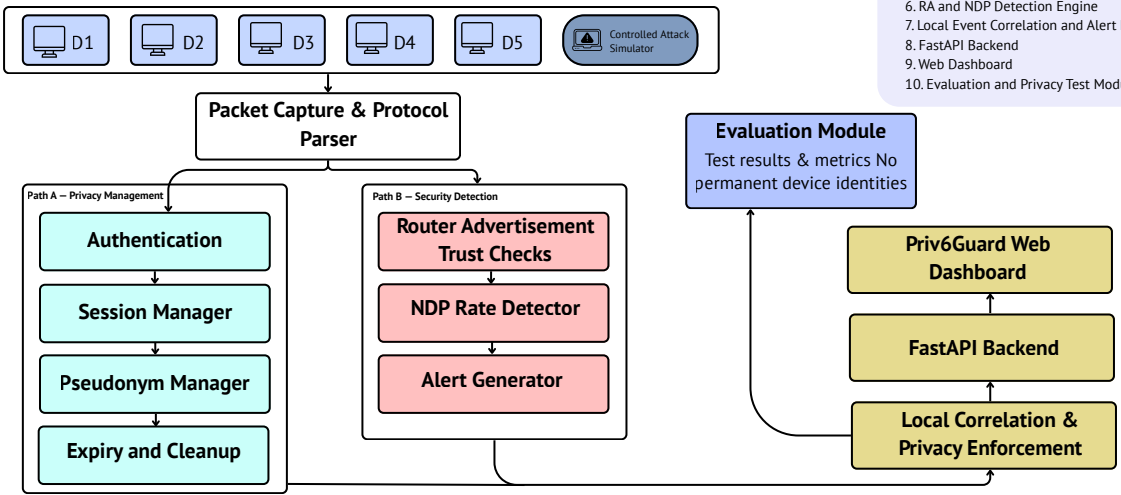
Key Enhancements

- Session-aware correlation across permitted address changes.
- Automatic pseudonym expiry and mapping cleanup.
- Local-only event correlation with tenant isolation.
- Configurable detection thresholds based on normal traffic.
- Repeatable attack simulations and measurable evaluation.
- Separation of security alerts from permanent device identity.

Important: The conventional column represents a comparison baseline to be implemented or measured. Actual improvements will be reported only after testing.

Solution Architecture & Design

1. IPv6 Device Simulator
2. Address Rotation Manager
3. Authentication & Session Manager
4. Temporary Pseudonym Manager
5. IPv6 Packet Capture and Parser
6. RA and NDP Detection Engine
7. Local Event Correlation and Alert Manager
8. FastAPI Backend
9. Web Dashboard
10. Evaluation and Privacy Test Module



Data Flow Diagram

Step 1: Device Enrollment

Five emulated IoT devices are enrolled and authenticated within the isolated test environment.

Step 2: Address Rotation

The simulator changes temporary IPv6 addresses according to the test schedule.

Step 3: Session & Pseudonym Management

The system uses an authenticated session to permit limited local correlation across address changes. Pseudonyms expire automatically.

Step 4: Packet Monitoring

IPv6 traffic is captured and relevant ICMPv6/NDP messages are classified.

Step 5: Threat Detection

- *RA events are checked against the configured trusted-router policy.*
- *NDP rates are compared with configured thresholds or baselines.*

Step 6: Local Correlation

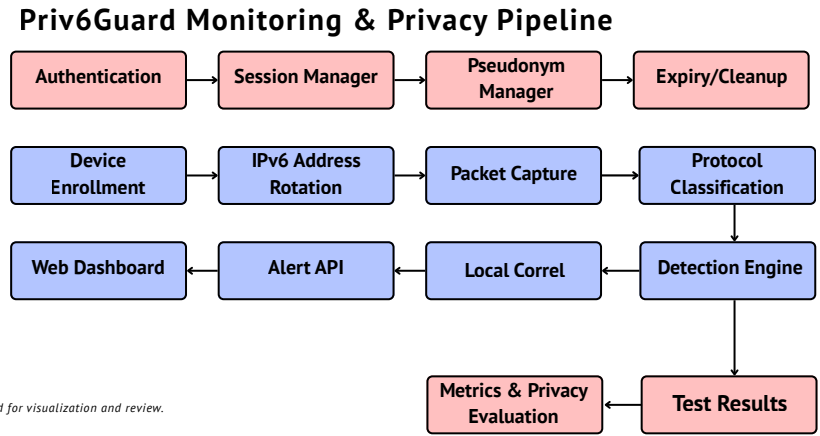
Permitted events are associated with active, short-lived local sessions. Expired mappings are not used for further correlation.

Step 7: Alert & Visualization

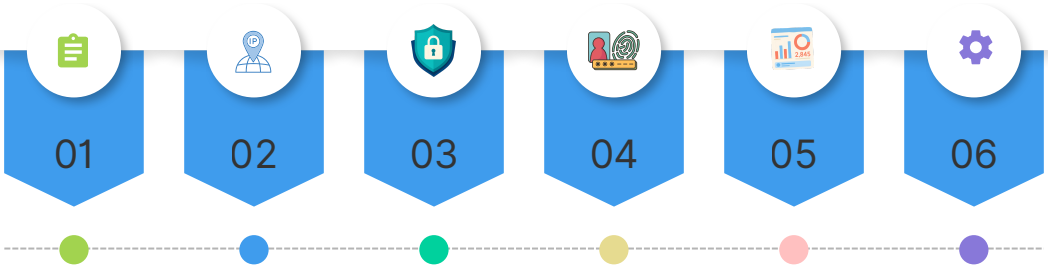
The backend sends event summaries to the dashboard for visualization and review.

Step 8: Evaluation

Test results are used to calculate detection performance, false positives, latency, pseudonym-expiry behaviour, and tenant isolation.



Deployment Timeline



Research & Setup

- Study RFC 8981, RFC 4861, and relevant IPv6 security concepts.
- Configure the development environment.
- Create the private GitHub repository.

IPv6 Simulation

- Build one isolated IPv6 LAN.
- Configure five emulated devices.
- Implement and verify temporary-address rotation.

Privacy & Sessions

- Implement device authentication.
- Create short-lived pseudonyms and session management.
- Test expiration, cleanup, and tenant isolation.

Security Monitoring

- Implement Scapy packet capture and NDP classification.
- Build rogue RA trust-policy detection.
- Implement NDP-rate anomaly detection.
- Validate both attacks in the isolated test network.

Dashboard Integration

- Build the React dashboard.
- Integrate FastAPI endpoints and live event updates.
- Display device sessions, rotation events, alerts, and metrics.

Evaluation & Submission

- Run normal-traffic and attack scenarios.
- Measure detection rate, false positives, and latency.
- Document privacy limitations and test results.
- Prepare the final report, screenshots, and demonstration.

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Thank You for Your Attention!